

RESPONSE TO THE REFEREE: NEURAL BELLMAN OPERATORS

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Revision R59 responds to the 9 October 2026 advisory report on R54. The original Neural Bellman Operators paper retains its constructive and economic targets and adds lossless trained-neural action search, prospective transcript invariance and complete matched-work evidence from the separately frozen R58 execution.

1. RESPONSE TO THE REFEREE

1.1. *Neural Bellman Operators* — Revision R59

We thank the referee for identifying the need to connect a valid policy certificate to a specific candidate computation, a prospective economic return and the complete work required to obtain that return. This revision continues the original **Neural Bellman Operators** paper. Its title, controlled economic laws, constructive own-future backend, feasible action witnesses, primitive resource account and original Bellman-accuracy objective are retained.

The controlling report is the 9 October 2026 advisory review on commit adf1256cff9cde365246a3db2dac90c72fda3b13, assessing R54 manuscript 00e837adb431f4d4b5248fc6d5ff1fc927dd3b65. It is not a decision of the Econometric Society. The repository subsequently completed the R56 and R57 manuscripts and a separately frozen R58 scientific execution. R59 starts from R58 development commit 4b5f1cc3241fe8df103844f225913d91c387795b, whose ordinary main article still contained the R57 exposition. The revision integrates that later execution, supplies its lossless-search and prospective-invariance proofs, and reconstructs the complete scientific record. It does not rename already executed services as new R59 observations.

The new main-article sections are **Lossless search for trained neural action witnesses** and **Equal-policy work at prospective economic targets**. They extend,

rather than replace, the original constructive operator and the fitted-witness transfer theory. The supplement gives every timing repetition, complete service record, component clock, activation-reduction count and datewise accuracy account. The original seven-document family is retained: article, technical supplement, referee response and four complete/development companions.

The main new conclusion is precise. A lossless implementation of the trained-neural action solver returns the same exact witness and, under the stated controller premises, the same acquired policy, verification endpoints and prospective stopping history as the algebraic implementation. The 36-service matched execution identifies a reduction in recorded complete-process median work at the same economic and statistical accuracy. This is not a claim that identical policies have different welfare, nor that a general safe-improvement theorem belongs exclusively to neural representations.

1.2. *M1 and B1: The original neural computation and the role of safe improvement*

We retain the original NBO title and research object rather than replace the manuscript with a differently titled safe-policy-improvement paper. The reason is substantive: the article now distinguishes the constructive operator, a trained-neural action computation, its lossless implementation, and the representation-neutral verification rule that makes the returned action economically admissible.

The constructive backend still produces an own-future continuation and feasible witnesses from primitive moduli, with an explicit error and resource account against the original Bellman optimum. The trained backend is not merely a ReLU encoding of that min-plus construction. Its stored affine-ReLU continuation has analytically integrated innovation expectations and an exact action-lattice minimizer obtained by rational hinge partition and cubic-root isolation. The preceding R57 execution actually deployed that minimizer and retained common-only, menu-neural, exact-quadratic and tree-based controls.

The new theorem reduces the work of computing precisely that trained-neural witness. Features whose entire action range lies in one integrated activation region are reduced exactly to polynomial coefficients or zero. Remaining lattice objec-

tives are enclosed outward; only points proved unable to minimize are excluded. Surviving values are compared rationally with the original tie rule. The proof permits negative fitted weights and does not assume a convex fit. It also establishes a survivor-gap bound, a conditional output-cardinality bound, exact fallback behavior and explicit arithmetic/storage costs.

The interval-screening principle itself is representation neutral. The NBO-specific implementation is the exact reduction and evaluation of integrated trained ridges and its preservation of the original action witness. Likewise, the incumbent-protection theorem remains representation neutral and available to conventional generators. The manuscript does not infer a universal neural advantage from either principle.

1.3. *M2 and B4: A prospective stopping algorithm, not a retrospective catalogue*

The earlier R54 catalogue remains historical evidence and is not relabeled as a prospective service. The subsequent R55/R57 protocols introduced actual first-attainment controllers, and the R58 matched experiment follows a separately frozen prospective rule. Every solver/task/target/repetition service begins from primitives and pays for its own fit, verification, inference and outputs.

The targets require ten- or twenty-percent expected-cost reductions from installed zero investment. The estimand is the signed contrast $J(\text{candidate}) - q \backslash J(\text{installed})$, with $q=9/10$ or $4/5$, rather than a ratio of two estimated means. At each construction attempt, the complete candidate is frozen before inference. A nonpositive upper endpoint returns immediately. A positive lower endpoint advances the construction attempt; an unresolved interval advances the next declared inference look. Final failure returns a safe policy with budget-exhausted status, not a claimed successful or impossible target.

The R58 design has two declared construction attempts, three cumulative inference looks and 36 separate complete processes. Its finite family covers all possible cost, target-contrast and gain intervals. The audit reconstructs every first-attainment decision and verifies that no look or attempt was performed after the corresponding

stopping condition. Identical policies and streams yield identical stopping histories in the two solver modes.

The new transcript-invariance theorem states its premise explicitly: controller decisions depend on mathematical outputs, not elapsed time or solver diagnostics. A wall-clock deadline is a different contract and need not produce the same stopping history. Intermediate construction attempts are not deployed; all are verified against the same smooth installed reference, not against a silently differentiated discontinuous updated actor.

1.4. M3 and B2: Matched economic accuracy and the full computational bill

The new comparison matches more than a nominal target. Corresponding modes have identical fitted parameters, acquired partitions, action witnesses, selected policies, whole-cell error accounts, interval-path endpoints and statistical decisions. Their policy-cost difference is therefore exactly zero under every admissible initial law. This is stronger than an unresolved paired confidence interval, and it does not require a new paired sampling service.

The complete recorded service includes process invocation, imports, scientific-source checks, independently repeated fitting, every attempted action search, verification, own inference, serialization, process exit and log flush. The parent controller's later checkpoint write is separately identified and is not described as zero overhead. The supplement distinguishes component medians from complete-process medians and reports all output bytes and resident-memory peaks.

Thirty of the 36 services attain the declared target. Six exhaust the budget at the eight-state, six-date twenty-percent target. All are retained. The complete-process algebraic-to-screened median ratios lie between approximately 1.028 and 1.101 across the six task-target groups. The unsuccessful group is explicitly a comparison of complete exhaustion work, not successful first-attainment work. All three timing repetitions are printed rather than selecting the best run.

The R57 independent neural-versus-conventional cost comparisons remain the evidence for those distinct policies. Equality between the two lossless implementa-

tions is not transferred to other generators, and the new within-neural work gain is not used to reverse the inherited conventional ranking.

1.5. *M4 and B3: Error decomposition and a targeted removable cost*

The fitted-witness transfer theorem already separates center optimization loss, within-cell action-contrast variation, action-relevant continuation error, continuous-to-robust-lattice coverage and verification overshoot. Its explicit neural moduli and the directly computed continuous-action cover remain in the main article. Exact fitted minimization removes only the first error term.

R59 targets the computational cost of obtaining that zero optimization error without changing the remaining error budget. The lossless solver returns exactly the original algebraic witness, so its fitted optimization error, acquired feasibility, verified action gap and propagated Bellman-loss account are unchanged. A tighter numerical stopping condition is not smuggled into one mode to create a favorable runtime comparison.

The recorded datewise quantities include U-L, U-C and C-L, selected actions, extra-witness changes and the baseline policy without the added exact candidate. These quantities add cellwise; their separately displayed maxima need not add. The new tables preserve this convention. The all-state policy-loss account includes the installed-reference continuation term and is not replaced with the local greedy gap.

The new survivor bound quantifies the effect of endpoint width on exact-comparison work. Under an additional discrete quadratic-growth condition it also bounds the number of survivors. The condition is stated, proved and tested as a conditional result; a global curvature constant is not inferred merely because this catalogue happened to leave one survivor per query. The paper does not claim to have performed an unexecuted six-factor causal error ablation.

1.6. *M5 and B5: Non-tensor multidimensional execution and explicit scaling costs*

The retained and new prospective services use the original nonlinear cyclic investment family at dimension/horizon pairs (2,2), (4,4) and (8,6). Their observations use locally split binary partitions with 32 or 64 leaves, not unequal-spaced tensor grids

presented as non-tensor. Robust capacities and downward action indices are valid on entire leaves. The signed smooth-reference verifier and its dimension/horizon work account remain common to all candidate generators.

R59 adds a distinct work account for trained-neural action recovery. For width m , dimension d , N action-lattice points, m_c crossing ridges and s survivors, coefficient formation/reduction requires $O(dm)$ arithmetic operations; interval screening requires $O(N(1+m_c))$ interval operations; exact original-objective comparison requires $O(sm)$ rational operations. The vectorized implementation uses $O(N(1+m_c))$ interval working storage. Rational bit costs are explicitly separate.

These bounds do not erase dimension, horizon, acquisition or lattice dependence. Nor is the new screen claimed to be universally faster: valid wide intervals may retain the whole lattice, and a fallback pays for the original exact solver. The recorded multidimensional executions and operation counts are genuine; different task runners do not establish a hardware-controlled asymptotic scaling law. The original stronger Bellman-accuracy target remains in the theoretical and returned-policy error account.

1.7. *M6 and B6: Strong controls and an attributable intervention*

The manuscript retains the R57 five-mode experiment: common-only, ReLU-menu, ReLU-exact, quadratic-exact and ExtraTrees-menu. Common-only legitimately incurs no fictional fitting cost. Every mode receives the same exact terminal analysis and critic-independent signed verifier. Matched ReLU-menu/exact parameters isolate the added witness, while the raw nested-candidate record attributes strict endpoint and selected-action changes to that addition.

The R58 experiment answers the next question within that already executed neural generator: can the same exact witness be returned with less work? It changes only the nonterminal solver. Each mode independently reconstructs the fit, and the audit checks the resulting parameters rather than assuming identity from seeds. The same actions enter the same verifier in the same order. This prevents a change of verifier, fit, target or policy from being credited to the screening algorithm.

The observed reduction in exact comparisons has a complete accounting counterpart: all lattice points evaluated by the interval screen are counted. Terminal root solving is identical and excluded from the reported neural intervention counts. The earlier evidence that the common menu determines many actions is preserved; the paper does not assign the whole verifier's economic gain to the neural proposal.

1.8. *M7 and B7: Fitted approximate structure and transition uncertainty*

The complete sample-split learned-direction theorem and R55 structural-uncertainty execution are retained from R56/R57. Directions are fitted on construction data; independent validation forms a confidence box for uncertain action exposure; the same confidence box qualifies the nuisance allowance and the remaining action comparison. The corrected gate does not obtain an unknown true exposure from the simulator as a free verification oracle.

The maintained uncertainty model has a known nonlinear drift and bounded action-invariant innovation structure with uncertain exposure. The manuscript states this scope rather than generalizing it to an arbitrary unknown transition kernel. The original fitted-direction result is also kept distinct from the earlier deterministic hand-perturbation catalogue, including their different harmful plug-in counts and statistical meanings.

The new lossless search concerns a fixed stored fitted objective and therefore does not add an estimation assumption to that structural procedure. It preserves the exact witness supplied to any separately valid uncertainty-qualified verifier. It does not turn a fitted approximately null direction into an exact identity or waive the confidence-set allowance.

1.9. *M8 and B8: Controlled repetition versus independent learned objects*

The R57 experiment contains three distinct fitted seed realizations, checked by parameter identity. Its timing variability mixes changes in learned objects and computation. R58 instead deliberately repeats one fixed fitted workload and its path streams three times, isolating repeated work measurements. The two kinds of evidence are separately identified and are not pooled into a larger statistical sample.

Within each task, the R58 processes run sequentially on one available core with single-thread numerical libraries. Solver order alternates by target and repetition. The full environment, governor availability, memory, serialized bytes and complete process clocks are retained. Frequency was not controlled, and the paper says so. All three values and their medians appear in the supplement.

The audit reconstructs 2,304 distinct solver executions across the six saved model/verification inputs, including both algorithms and common terminal calls. There are 864 distinct nonterminal neural queries. It regenerates nine distinct stopping streams comprising 380,928 path rows, checks 132 recorded inference looks and 396 cost/contrast/gain interval records, and verifies 9,216 cell-date decisions across the full service catalogue. Memoized repeated calculations are checked by input and output identities, not counted as new independent observations. These are finite, reproducible workload results, not a population guarantee over economic tasks or hardware.

1.10. *M9 and B9: An economic consequence at an unchanged answer*

The executed targets concern actual expected cost under the original continuous initial and innovation laws. The positive gain intervals relative to the installed controller remain the basis for adoption decisions, including the eight-dimensional case whose stronger twenty-percent target is not certified at the available budget.

The new implementation comparison has an especially direct economic interpretation. When both solvers return exactly the same policy, its economic cost cancels from their total-cost comparison. For specified computation charges and a non-negative computation price, the total-cost difference is exactly the price times the charge difference. Relative to installation, the same gain interval is shifted by the implementation charge and installation fee. A lower specified computational charge expands the supported adoption region without inventing a difference in policy welfare.

The main article states and proves this result and reports the complete equal-policy work measurements that make it operational in the finite study. Recorded medians do not provide a distribution-free expected-runtime estimate, and no exter-

nal computation-price or installation-fee calibration is claimed. The contribution is a constructive and executable method for obtaining a verified economic answer more economically on the recorded matched workloads, together with its exact policy and stopping invariance; it is not an asserted universal inability of conventional methods to obtain that answer.

1.11. *M10 and B10: A coherent original paper with preserved development*

The active manuscript retains the connected NBO route: own-future operator, constructive continuation and primitive resources, trained neural action witness, fitted-witness transfer, acquired-state verification, prospective policy return and economic cost. R59 inserts the lossless computation result and its matched execution into this route. It neither retitles the paper nor replaces its economic model or Bellman objective.

Every earlier active section remains available, and the prior main article, technical supplement and response are preserved in a named snapshot. The complete/development companions retain the broader applications, original proofs, failed attempts and adverse numerical evidence under their own assumptions. The new expected-cost result is not silently extended to nonlinear recursive utility or games. Source and label-preservation checks enforce the distinction between reorganizing a reading path and discarding substantive content.

The new theory is proved in the main article. Detailed repeated clocks, every service return, operation counts and numerical semantics are in the supplement. Ordinary sources, frozen protocols, raw arrays, exact replay and final manifests permit another referee to inspect the claims without relying on a favorable summary or expiring workflow output.

1.12. *Verification and delivery*

The publication reconstructs both exact solvers from each distinct saved fitted input, regenerates every required whole-cell certificate and original-law interval path, verifies stored inference using exact-rational moments, and replays every prospective stop. Every frozen file, process prefix and parameter/policy identity is

checked. The prior audit and test chain is retained alongside the R58 solver fixtures
and 12 new exact theorem regressions.

The seven documents are compiled from ordinary sources, with reference, label,
glyph and overflow checks. A clean archive removes generated documents and
tables and reconstructs them after verifying the scientific-source/evidence bind-
ings. The final delivery manifest binds the tested source and published package.
Publication adds zero independently trained objects, zero fresh policy-cost samples
and no replacement performance clocks. These checks support reproducibility and
internal consistency; they do not presume the outcome of renewed mathematical or
editorial review.